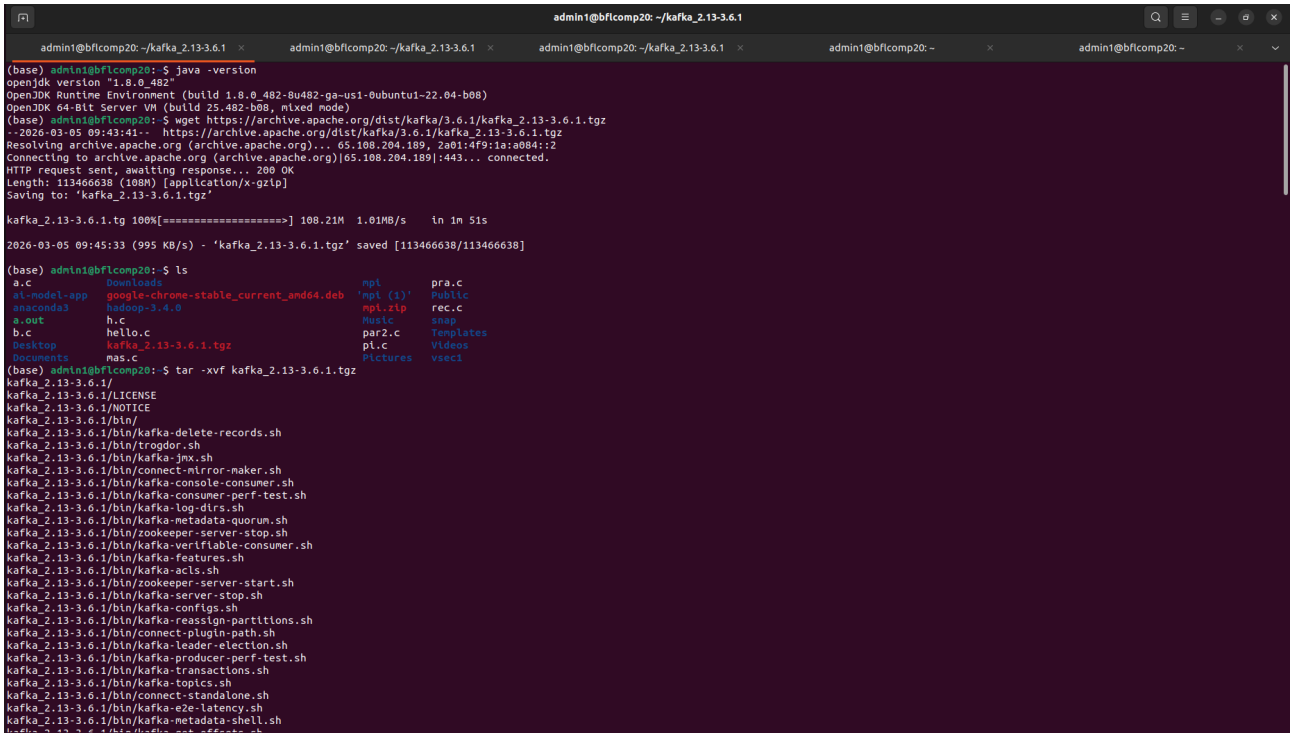
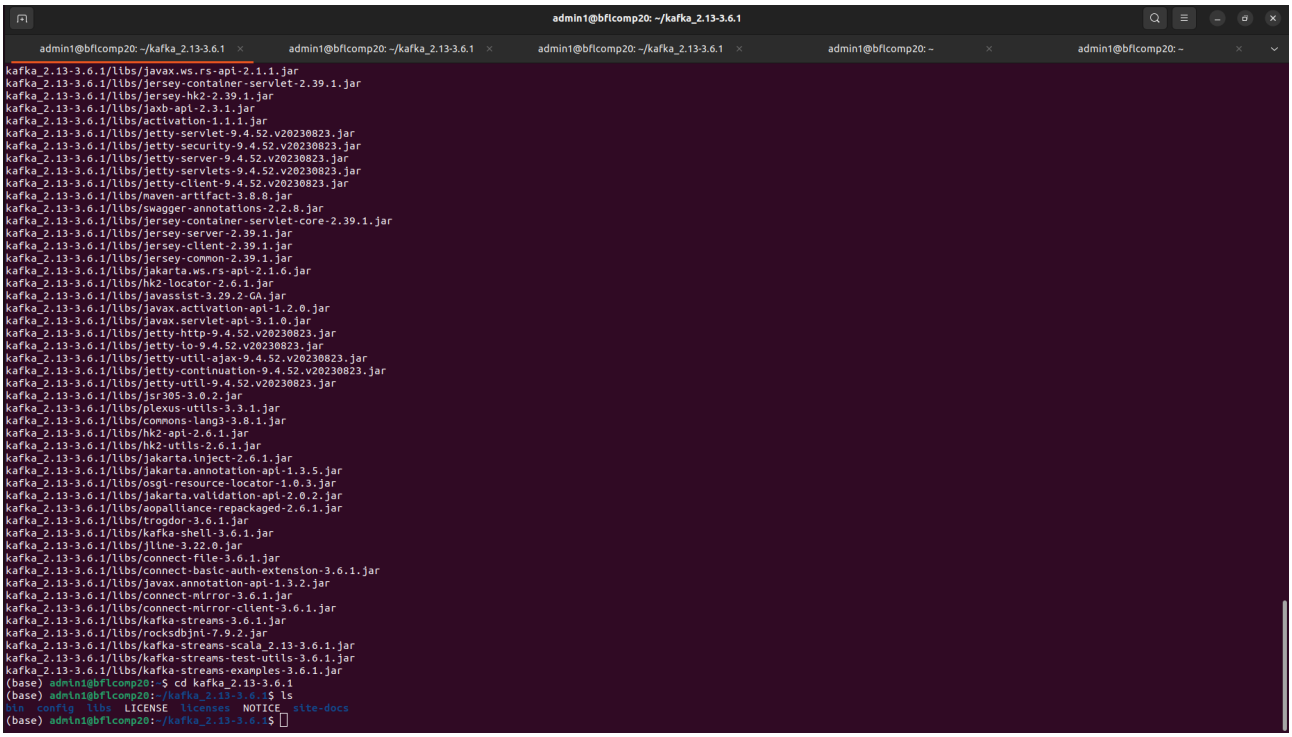
ASSIGNMENT – V

TITLE :
Create a real-time fraud detection system using streaming data from financial transactions.

NAME : Shinde Shubham Dnyandev. **ROLL NO :** 23107121. **BATCH :** B





admin1@bflcomp20: ~/kafka_2.13-3.6.1

admin1@bflcomp20: ~/kafka_2.13-3.6.1 | admin1@bflcomp20: ~/kafka_2.13-3.6.1 | admin1@bflcomp20: ~/kafka_2.13-3.6.1 | admin1@bflcomp20: ~ | admin1@bflcomp20: ~

```
(base) admin1@bflcomp20:~/kafka_2.13-3.6.1$ bin/zookeeper-server-start.sh config/zookeeper.properties
[2026-03-05 09:46:13,820] INFO Reading configuration from: config/zookeeper.properties (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2026-03-05 09:46:13,820] WARN config/zookeeper.properties is relative. Prepend ./ to indicate that you're sure! (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2026-03-05 09:46:13,822] INFO clientPortAddress is 0.0.0.0:2181 (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2026-03-05 09:46:13,822] INFO secureClientPort is not set (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2026-03-05 09:46:13,822] INFO observerMasterPort is not set (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2026-03-05 09:46:13,822] INFO metricsProvider.className is org.apache.zookeeper.metrics.impl.DefaultMetricsProvider (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2026-03-05 09:46:13,823] INFO autopurge.snapRetainCount set to 3 (org.apache.zookeeper.server.DatadirCleanupManager)
[2026-03-05 09:46:13,823] INFO autopurge.purgeInterval set to 0 (org.apache.zookeeper.server.DatadirCleanupManager)
[2026-03-05 09:46:13,823] INFO Purge task is not scheduled. (org.apache.zookeeper.server.DatadirCleanupManager)
[2026-03-05 09:46:13,823] WARN Either no config or no quorum defined in config, running in standalone mode (org.apache.zookeeper.server.quorum.QuorumPeerMain)
[2026-03-05 09:46:13,824] INFO Log4j 1.2 jmx support not found; jmx disabled. (org.apache.zookeeper.jmx.ManagedUtil)
[2026-03-05 09:46:13,824] INFO Reading configuration from: config/zookeeper.properties (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2026-03-05 09:46:13,824] WARN config/zookeeper.properties is relative. Prepend ./ to indicate that you're sure! (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2026-03-05 09:46:13,824] INFO clientPortAddress is 0.0.0.0:2181 (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2026-03-05 09:46:13,824] INFO secureClientPort is not set (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2026-03-05 09:46:13,824] INFO observerMasterPort is not set (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2026-03-05 09:46:13,824] INFO metricsProvider.className is org.apache.zookeeper.metrics.impl.DefaultMetricsProvider (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2026-03-05 09:46:13,824] INFO Starting server (org.apache.zookeeper.server.ZooKeeperServerMain)
[2026-03-05 09:46:13,831] INFO ServerMetrics initialized with provider org.apache.zookeeper.metrics.impl.DefaultMetricsProvider@b97c004 (org.apache.zookeeper.server.ServerMetrics)
[2026-03-05 09:46:13,833] INFO ACL digest algorithm is: SHA1 (org.apache.zookeeper.server.auth.DigestAuthenticationProvider)
[2026-03-05 09:46:13,833] INFO zookeeper.DigestAuthenticationProvider.enabled = true (org.apache.zookeeper.server.auth.DigestAuthenticationProvider)
[2026-03-05 09:46:13,834] INFO zookeeper.snapshot.trust.empty : false (org.apache.zookeeper.server.persistence.FileTxnSnapLog)
[2026-03-05 09:46:13,839] INFO  (org.apache.zookeeper.server.ZooKeeperServer)
[2026-03-05 09:46:13,839] INFO                                                     (org.apache.zookeeper.server.ZooKeeperServer)
[2026-03-05 09:46:13,839] INFO  |___  /                 | |                        (org.apache.zookeeper.server.ZooKeeperServer)
[2026-03-05 09:46:13,839] INFO     / /   ___   ___   ___ | |  __   ___   ___  _ __   (org.apache.zookeeper.server.ZooKeeperServer)
[2026-03-05 09:46:13,839] INFO    / /   / __ \ / __ \ / _ \| | / /  / _ \ / _ \ | '_ \ | '_ \ (org.apache.zookeeper.server.ZooKeeperServer)
[2026-03-05 09:46:13,839] INFO   / /__ | (_) ) (_) ) (_) )  <  / /__ | (_) ) (_) ) | | ) | | ) | (org.apache.zookeeper.server.ZooKeeperServer)
[2026-03-05 09:46:13,839] INFO  /_____| \____/ \____/ \___/|_|\_\ \___| \____/ \___/ |_| |_| |_| (org.apache.zookeeper.server.ZooKeeperServer)
[2026-03-05 09:46:13,840] INFO                                                     | |                (org.apache.zookeeper.server.ZooKeeperServer)
[2026-03-05 09:46:13,840] INFO                                                     |_|                (org.apache.zookeeper.server.ZooKeeperServer)
[2026-03-05 09:46:13,840] INFO  (org.apache.zookeeper.server.ZooKeeperServer)
[2026-03-05 09:46:13,840] INFO Server environment:zookeeper.version=3.8.3-6ad6d364c7c0bcf0de452d54ebefa3058098ab56, built on 2023-10-05 10:34 UTC (org.apache.zookeeper.server.ZooKeeperServer)
[2026-03-05 09:46:13,840] INFO Server environment:host.name=bflcomp20 (org.apache.zookeeper.server.ZooKeeperServer)
[2026-03-05 09:46:13,840] INFO Server environment:java.version=1.8.0_482 (org.apache.zookeeper.server.ZooKeeperServer)
[2026-03-05 09:46:13,840] INFO Server environment:java.vendor=Private Build (org.apache.zookeeper.server.ZooKeeperServer)
[2026-03-05 09:46:13,840] INFO Server environment:java.home=/usr/lib/jvm/java-8-openjdk-amd64/jre (org.apache.zookeeper.server.ZooKeeperServer)
[2026-03-05 09:46:13,840] INFO Server environment:java.class.path=/home/admin1/kafka_2.13-3.6.1/bin/../libs/activation-1.1.1.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/aopalliance-repackaged-2.6.1.jar:
/home/admin1/kafka_2.13-3.6.1/bin/../libs/argparse4j-0.7.0.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/audience-annotations-0.12.0.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/caffeine-2.9.3.jar:/home/
admin1/kafka_2.13-3.6.1/bin/../libs/checker-qual-3.19.0.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/commons-beanutils-1.9.4.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/commons-cli-1.4.jar:/home/admin1
/kafka_2.13-3.6.1/bin/../libs/commons-collections-3.2.2.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/commons-digester-2.1.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/commons-io-2.11.0.jar:/home/admin1/
kafka_2.13-3.6.1/bin/../libs/commons-lang3-3.8.1.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/commons-logging-1.2.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/commons-validator-1.7.jar:/home/admin1/kafk
a_2.13-3.6.1/bin/../libs/connect-api-3.6.1.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/connect-basic-auth-extension-3.6.1.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/connect-json-3.6.1.jar:/home/admin
1/kafka_2.13-3.6.1/bin/../libs/connect-mirror-3.6.1.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/connect-mirror-client-3.6.1.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/connect-runtime-3.6.1.jar:/home/
admin1/kafka_2.13-3.6.1/bin/../libs/connect-transforms-3.6.1.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/error_prone_annotations-2.10.0.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/hk2-api-2.6.1.jar:/h
ome/admin1/kafka_2.13-3.6.1/bin/../libs/hk2-locator-2.6.1.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/hk2-utils-2.6.1.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/jackson-annotations-2.13.5.jar:/home/a
dmin1/kafka_2.13-3.6.1/bin/../libs/jackson-core-2.13.5.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/jackson-databind-2.13.5.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/jackson-dataformat-csv-2.13.5.jar
:/home/admin1/kafka_2.13-3.6.1/bin/../libs/jackson-datatype-jdk8-2.13.5.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/jackson-jaxrs-base-2.13.5.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/jackson-jaxrs-
json-provider-2.13.5.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/jackson-module-jaxb-annotations-2.13.5.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/jackson-module-scala_2.13-2.13.5.jar:/home/admin1/ka
fka_2.13-3.6.1/bin/../libs/jakarta.activation-api-1.2.2.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/jakarta.annotation-api-1.3.5.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/jakarta.inject-2.6.1.jar:/h
ome/admin1/kafka_2.13-3.6.1/bin/../libs/jakarta.validation-api-2.0.2.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/jakarta.ws.rs-api-2.1.6.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/jakarta.xml.bind-ap
i-2.3.3.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/javassist-3.29.2-GA.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/javax.activation-api-1.2.0.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/javax.annot
```

admin1@bflcomp20: ~/kafka_2.13-3.6.1

admin1@bflcomp20: ~/kafka_2.13-3.6.1 | admin1@bflcomp20: ~/kafka_2.13-3.6.1 | admin1@bflcomp20: ~/kafka_2.13-3.6.1 | admin1@bflcomp20: ~ | admin1@bflcomp20: ~

```
(base) admin1@bflcomp20:~/kafka_2.13-3.6.1$ bin/kafka-server-start.sh config/server.properties
[2026-03-05 09:46:33,801] INFO Registered kafka:type=kafka.Log4jController MBean (kafka.utils.Log4jControllerRegistration$)
[2026-03-05 09:46:33,948] INFO Setting -D jdk.tls.rejectClientInitiatedRenegotiation=true to disable client-initiated TLS renegotiation (org.apache.zookeeper.common.X509Util)
[2026-03-05 09:46:33,992] INFO Registered signal handlers for TERM, INT, HUP (org.apache.kafka.common.utils.LoggingSignalHandler)
[2026-03-05 09:46:33,993] INFO starting (kafka.server.KafkaServer)
[2026-03-05 09:46:33,993] INFO Connecting to zookeeper on localhost:2181 (kafka.server.KafkaServer)
[2026-03-05 09:46:34,000] INFO [ZooKeeperClient Kafka server] Initializing a new session to localhost:2181. (kafka.zookeeper.ZooKeeperClient)
[2026-03-05 09:46:34,002] INFO Client environment:zookeeper.version=3.8.3-6ad6d364c7c0bcf0de452d54ebefa3058098ab56, built on 2023-10-05 10:34 UTC (org.apache.zookeeper.ZooKeeper)
[2026-03-05 09:46:34,002] INFO Client environment:host.name=bflcomp20 (org.apache.zookeeper.ZooKeeper)
[2026-03-05 09:46:34,002] INFO Client environment:java.version=1.8.0_482 (org.apache.zookeeper.ZooKeeper)
[2026-03-05 09:46:34,002] INFO Client environment:java.vendor=Private Build (org.apache.zookeeper.ZooKeeper)
[2026-03-05 09:46:34,002] INFO Client environment:java.home=/usr/lib/jvm/java-8-openjdk-amd64/jre (org.apache.zookeeper.ZooKeeper)
[2026-03-05 09:46:34,002] INFO Client environment:java.class.path=/home/admin1/kafka_2.13-3.6.1/bin/../libs/activation-1.1.1.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/aopalliance-repackaged-2.6.1.jar:
/home/admin1/kafka_2.13-3.6.1/bin/../libs/argparse4j-0.7.0.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/audience-annotations-0.12.0.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/caffeine-2.9.3.jar:/home/
admin1/kafka_2.13-3.6.1/bin/../libs/checker-qual-3.19.0.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/commons-beanutils-1.9.4.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/commons-cli-1.4.jar:/home/admin1
/kafka_2.13-3.6.1/bin/../libs/commons-collections-3.2.2.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/commons-digester-2.1.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/commons-io-2.11.0.jar:/home/admin1/
kafka_2.13-3.6.1/bin/../libs/commons-lang3-3.8.1.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/commons-logging-1.2.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/commons-validator-1.7.jar:/home/admin1/kafk
a_2.13-3.6.1/bin/../libs/connect-api-3.6.1.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/connect-basic-auth-extension-3.6.1.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/connect-json-3.6.1.jar:/home/admin
1/kafka_2.13-3.6.1/bin/../libs/connect-mirror-3.6.1.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/connect-mirror-client-3.6.1.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/connect-runtime-3.6.1.jar:/home/
admin1/kafka_2.13-3.6.1/bin/../libs/connect-transforms-3.6.1.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/error_prone_annotations-2.10.0.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/hk2-api-2.6.1.jar:/h
ome/admin1/kafka_2.13-3.6.1/bin/../libs/hk2-locator-2.6.1.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/hk2-utils-2.6.1.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/jackson-annotations-2.13.5.jar:/home/a
dmin1/kafka_2.13-3.6.1/bin/../libs/jackson-core-2.13.5.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/jackson-databind-2.13.5.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/jackson-dataformat-csv-2.13.5.jar
:/home/admin1/kafka_2.13-3.6.1/bin/../libs/jackson-datatype-jdk8-2.13.5.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/jackson-jaxrs-base-2.13.5.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/jackson-jaxrs-
json-provider-2.13.5.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/jackson-module-jaxb-annotations-2.13.5.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/jackson-module-scala_2.13-2.13.5.jar:/home/admin1/ka
fka_2.13-3.6.1/bin/../libs/jakarta.activation-api-1.2.2.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/jakarta.annotation-api-1.3.5.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/jakarta.inject-2.6.1.jar:/h
ome/admin1/kafka_2.13-3.6.1/bin/../libs/jakarta.validation-api-2.0.2.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/jakarta.ws.rs-api-2.1.6.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/jakarta.xml.bind-ap
i-2.3.3.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/javassist-3.29.2-GA.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/javax.activation-api-1.2.0.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/javax.annot
ation-api-1.3.2.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/javax.servlet-api-3.1.0.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/javax.ws.rs-api-2.1.1.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/jaxb
-api-2.3.1.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/jersey-client-2.39.1.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/jersey-common-2.39.1.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/jersey-contai
ner-servlet-2.39.1.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/jersey-container-servlet-core-2.39.1.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/jersey-hk2-2.39.1.jar:/home/admin1/kafka_2.13-3.6.1/bin/
../libs/jersey-server-2.39.1.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/jetty-client-9.4.52.v20230823.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/jetty-continuation-9.4.52.v20230823.jar:/home/admin1/
kafka_2.13-3.6.1/bin/../libs/jetty-http-9.4.52.v20230823.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/jetty-io-9.4.52.v20230823.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/jetty-security-9.4.52.v202308
23.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/jetty-server-9.4.52.v20230823.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/jetty-servlet-9.4.52.v20230823.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/je
tty-servlets-9.4.52.v20230823.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/jetty-util-9.4.52.v20230823.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/jetty-util-ajax-9.4.52.v20230823.jar:/home/admin1/kafk
a_2.13-3.6.1/bin/../libs/jline-3.22.0.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/jopt-simple-5.0.4.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/jose4j-0.9.3.jar:/home/admin1/kafka_2.13-3.6.1/bin/../li
bs/jsr305-3.0.2.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/kafka_2.13-3.6.1.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/kafka-clients-3.6.1.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/kafka-group-c
oordinator-3.6.1.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/kafka-log4j-appender-3.6.1.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/kafka-metadata-3.6.1.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/k
afka-raft-3.6.1.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/kafka-server-common-3.6.1.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/kafka-shell-3.6.1.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/kafka-
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reams-examples-3.6.1.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/kafka-streams-scala_2.13-3.6.1.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/kafka-streams-test-utils-3.6.1.jar:/home/admin1/kafka_2.13-3
.6.1/bin/../libs/kafka-tools-3.6.1.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/kafka-tools-api-3.6.1.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/lz4-java-1.8.0.jar:/home/admin1/kafka_2.13-3.6.1/bin/..
/libs/maven-artifact-3.8.8.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/metrics-core-2.2.0.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/metrics-core-4.1.12.1.jar:/home/admin1/kafka_2.13-3.6.1/bin/../lib
s/netty-buffer-4.1.100.Final.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/netty-codec-4.1.100.Final.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/netty-common-4.1.100.Final.jar:/home/admin1/kafka_2.13-3.
6.1/bin/../libs/netty-handler-4.1.100.Final.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/netty-resolver-4.1.100.Final.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/netty-transport-4.1.100.Final.jar:/home
/admin1/kafka_2.13-3.6.1/bin/../libs/netty-transport-classes-epoll-4.1.100.Final.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/netty-transport-native-epoll-4.1.100.Final.jar:/home/admin1/kafka_2.13-3.6.1/
bin/../libs/netty-transport-native-unix-common-4.1.100.Final.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/osgi-resource-locator-1.0.3.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/paranamer-2.8.jar:/home
/admin1/kafka_2.13-3.6.1/bin/../libs/pcollections-4.0.1.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/plexus-utils-3.3.1.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/reflections-0.10.2.jar:/home/admin1/k
afka_2.13-3.6.1/bin/../libs/reload4j-1.2.25.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/rocksdbjni-7.9.2.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/scala-collection-compat_2.13-2.10.0.jar:/home/admin
1/kafka_2.13-3.6.1/bin/../libs/scala-java8-compat_2.13-1.0.2.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/scala-library-2.13.11.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/scala-logging_2.13-3.9.4.jar:
/home/admin1/kafka_2.13-3.6.1/bin/../libs/scala-reflect-2.13.11.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/slf4j-api-1.7.36.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/slf4j-reload4j-1.7.36.jar:/home
/admin1/kafka_2.13-3.6.1/bin/../libs/snappy-java-1.1.10.5.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/swagger-annotations-2.2.8.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/trogdor-3.6.1.jar:/home/admi
n1/kafka_2.13-3.6.1/bin/../libs/zookeeper-3.8.3.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/zookeeper-jute-3.8.3.jar:/home/admin1/kafka_2.13-3.6.1/bin/../libs/zstd-jni-1.5.5-1.jar (org.apache.zookeeper.
ZooKeeper)
```

admin1@bflcomp20: ~

admin1@bflcomp20: ~/kafka_2.13-3.6.1 | admin1@bflcomp20: ~/kafka_2.13-3.6.1 | admin1@bflcomp20: ~/kafka_2.13-3.6.1 | admin1@bflcomp20: ~ | admin1@bflcomp20: ~

```
(base) admin1@bflcomp20:~/kafka_2.13-3.6.1$ jps
8327 Jps
7770 Kafka
7293 QuorumPeerMain
(base) admin1@bflcomp20:~/kafka_2.13-3.6.1$ bin/kafka-topics.sh --create \
--topic transactions \
--bootstrap-server localhost:9092 \
--partitions 1 \
--replication-factor 1
Created topic transactions.
(base) admin1@bflcomp20:~/kafka_2.13-3.6.1$ bin/kafka-topics.sh --list --bootstrap-server localhost:9092
transactions
(base) admin1@bflcomp20:~/kafka_2.13-3.6.1$ cd ~
(base) admin1@bflcomp20:~$ pip install kafka-python pandas
Collecting kafka-python
  Downloading kafka_python-2.3.0-py2.py3-none-any.whl.metadata (10.0 kB)
Requirement already satisfied: pandas in ./anaconda3/lib/python3.9/site-packages (1.4.4)
Requirement already satisfied: python-dateutil>=2.8.1 in ./anaconda3/lib/python3.9/site-packages (from pandas) (2.8.2)
Requirement already satisfied: pytz>=2020.1 in ./anaconda3/lib/python3.9/site-packages (from pandas) (2022.1)
Requirement already satisfied: numpy>=1.18.5 in ./anaconda3/lib/python3.9/site-packages (from pandas) (1.24.4)
Requirement already satisfied: six>=1.5 in ./anaconda3/lib/python3.9/site-packages (from python-dateutil>=2.8.1->pandas) (1.16.0)
Downloading kafka_python-2.3.0-py2.py3-none-any.whl (326 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 326.3/326.3 kB 5.2 MB/s eta 0:00:00
Installing collected packages: kafka-python
Successfully installed kafka-python-2.3.0

[notice] A new release of pip is available: 24.0 -> 26.0.1
[notice] To update, run: pip install --upgrade pip
(base) admin1@bflcomp20:~$ nano producer.py
(base) admin1@bflcomp20:~$ python3 producer.py
Sent: {'transaction_id': 9386, 'amount': 4041, 'is_fraud': 0}
Sent: {'transaction_id': 3262, 'amount': 4274, 'is_fraud': 0}
Sent: {'transaction_id': 1705, 'amount': 4052, 'is_fraud': 0}
Sent: {'transaction_id': 4556, 'amount': 4288, 'is_fraud': 0}
Sent: {'transaction_id': 8487, 'amount': 4953, 'is_fraud': 0}
Sent: {'transaction_id': 4467, 'amount': 4100, 'is_fraud': 0}
Sent: {'transaction_id': 7845, 'amount': 4151, 'is_fraud': 0}
Sent: {'transaction_id': 6013, 'amount': 4999, 'is_fraud': 0}
Sent: {'transaction_id': 2963, 'amount': 4083, 'is_fraud': 0}
Sent: {'transaction_id': 4922, 'amount': 4219, 'is_fraud': 0}
Sent: {'transaction_id': 1845, 'amount': 4877, 'is_fraud': 0}
Sent: {'transaction_id': 3860, 'amount': 4276, 'is_fraud': 0}
Sent: {'transaction_id': 2547, 'amount': 4574, 'is_fraud': 0}
Sent: {'transaction_id': 7610, 'amount': 3778, 'is_fraud': 0}
Sent: {'transaction_id': 9960, 'amount': 3519, 'is_fraud': 0}
Sent: {'transaction_id': 4744, 'amount': 4388, 'is_fraud': 0}
Sent: {'transaction_id': 9713, 'amount': 3839, 'is_fraud': 0}
Sent: {'transaction_id': 7385, 'amount': 4922, 'is_fraud': 0}
Sent: {'transaction_id': 8488, 'amount': 4860, 'is_fraud': 0}
Sent: {'transaction_id': 3652, 'amount': 3985, 'is_fraud': 0}
Sent: {'transaction_id': 5171, 'amount': 4251, 'is_fraud': 0}
Sent: {'transaction_id': 8562, 'amount': 4708, 'is_fraud': 0}
Sent: {'transaction_id': 5174, 'amount': 3686, 'is_fraud': 0}
```

admin1@bflcomp20: ~

GNU nano 6.2 producer.py

```python
from kafka import KafkaProducer
import json, time, random

producer = KafkaProducer(
    bootstrap_servers='localhost:9092',
    value_serializer=lambda v: json.dumps(v).encode('utf-8')
)

while True:
    transaction = {
        "transaction_id": random.randint(1000, 9999),
        "amount": random.randint(3500, 5000),
        "is_fraud": 0
    }

    producer.send('transactions', transaction)
    print("Sent:", transaction)
    time.sleep(1)
```

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark M-] To Bracket M-Q Previous ^B Back ^← Prev Word
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy ^Q Where Was M-W Next ^F Forward ^→ Next Word

admin1@bflcomp20: ~/spark-3.5.1-bin-hadoop3

admin1@bflcomp20: ~/kafka_2.13-3.6.1 | admin1@bflcomp20: ~/kafka_2.13-3.6.1 | admin1@bflcomp20: ~/kafka_2.13-3.6.1 | admin1@bflcomp20: ~ | admin1@bflcomp20: ~/spark-3.5.1-bin-had...

```
(base) admin1@bflcomp20:~$ wget https://archive.apache.org/dist/spark/spark-3.5.1/spark-3.5.1-bin-hadoop3.tgz
--2026-03-05 09:50:03--  https://archive.apache.org/dist/spark/spark-3.5.1/spark-3.5.1-bin-hadoop3.tgz
Resolving archive.apache.org (archive.apache.org)... 65.108.204.189, 2a01:4f9:1a:a084::2
Connecting to archive.apache.org (archive.apache.org)|65.108.204.189|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 400446614 (382M) [application/x-gzip]
Saving to: ‘spark-3.5.1-bin-hadoop3.tgz’

spark-3.5.1-bin-hadoop3.tgz                         23%[==========>                                                                                                  ]  89.50M  1.09MB/s    eta 6m
spark-3.5.1-bin-hadoop3.tgz                        100%[===================================================================================================>] 381.90M   992KB/s    in 6m 16s

2026-03-05 09:56:20 (1.02 MB/s) - ‘spark-3.5.1-bin-hadoop3.tgz’ saved [400446614/400446614]

(base) admin1@bflcomp20:~$ tar -xvf spark-3.5.1-bin-hadoop3.tgz
spark-3.5.1-bin-hadoop3/
spark-3.5.1-bin-hadoop3/sbin/
spark-3.5.1-bin-hadoop3/sbin/spark-config.sh
spark-3.5.1-bin-hadoop3/sbin/stop-slave.sh
spark-3.5.1-bin-hadoop3/sbin/stop-mesos-dispatcher.sh
spark-3.5.1-bin-hadoop3/sbin/start-workers.sh
spark-3.5.1-bin-hadoop3/sbin/start-slaves.sh
spark-3.5.1-bin-hadoop3/sbin/start-all.sh
spark-3.5.1-bin-hadoop3/sbin/stop-all.sh
spark-3.5.1-bin-hadoop3/sbin/workers.sh
spark-3.5.1-bin-hadoop3/sbin/start-mesos-dispatcher.sh
spark-3.5.1-bin-hadoop3/sbin/spark-daemon.sh
spark-3.5.1-bin-hadoop3/sbin/decommission-worker.sh
spark-3.5.1-bin-hadoop3/sbin/slaves.sh
spark-3.5.1-bin-hadoop3/sbin/stop-mesos-shuffle-service.sh
spark-3.5.1-bin-hadoop3/sbin/stop-history-server.sh
spark-3.5.1-bin-hadoop3/sbin/stop-worker.sh
spark-3.5.1-bin-hadoop3/sbin/decommission-slave.sh
spark-3.5.1-bin-hadoop3/sbin/stop-thriftserver.sh
spark-3.5.1-bin-hadoop3/sbin/start-worker.sh
spark-3.5.1-bin-hadoop3/sbin/stop-slaves.sh
spark-3.5.1-bin-hadoop3/sbin/start-connect-server.sh
spark-3.5.1-bin-hadoop3/sbin/start-thriftserver.sh
spark-3.5.1-bin-hadoop3/sbin/start-history-server.sh
spark-3.5.1-bin-hadoop3/sbin/stop-connect-server.sh
spark-3.5.1-bin-hadoop3/sbin/spark-daemons.sh
spark-3.5.1-bin-hadoop3/sbin/start-slave.sh
spark-3.5.1-bin-hadoop3/sbin/start-master.sh
spark-3.5.1-bin-hadoop3/sbin/start-mesos-shuffle-service.sh
spark-3.5.1-bin-hadoop3/sbin/stop-master.sh
spark-3.5.1-bin-hadoop3/sbin/stop-workers.sh
spark-3.5.1-bin-hadoop3/licenses/
spark-3.5.1-bin-hadoop3/licenses/LICENSE-automaton.txt
spark-3.5.1-bin-hadoop3/licenses/LICENSE-d3.min.js.txt
spark-3.5.1-bin-hadoop3/licenses/LICENSE-scopt.txt
spark-3.5.1-bin-hadoop3/licenses/LICENSE-leveldbjni.txt
spark-3.5.1-bin-hadoop3/licenses/LICENSE-blas.txt
spark-3.5.1-bin-hadoop3/licenses/LICENSE-spire.txt
spark-3.5.1-bin-hadoop3/licenses/LICENSE-paranamer.txt
```

admin1@bflcomp20: ~/spark-3.5.1-bin-hadoop3

admin1@bflcomp20: ~/kafka_2.13-3.6.1 | admin1@bflcomp20: ~/kafka_2.13-3.6.1 | admin1@bflcomp20: ~/kafka_2.13-3.6.1 | admin1@bflcomp20: ~ | admin1@bflcomp20: ~/spark-3.5.1-bin-had...

GNU nano 6.2 fraud_detector.py *

```python
from pyspark.sql import SparkSession
from pyspark.sql.functions import col, from_json
from pyspark.sql.types import StructType, IntegerType

# Create Spark session
spark = SparkSession.builder \
    .appName("FraudDetection") \
    .getOrCreate()

# Define schema
schema = StructType() \
    .add("transaction_id", IntegerType()) \
    .add("amount", IntegerType()) \
    .add("is_fraud", IntegerType())

# Read stream from Kafka
df = spark.readStream \
    .format("kafka") \
    .option("kafka.bootstrap.servers", "localhost:9092") \
    .option("subscribe", "transactions") \
    .option("startingOffsets", "earliest") \
    .load()

# Convert Kafka value to string
value_df = df.selectExpr("CAST(value AS STRING)")

# Convert JSON string to columns
json_df = value_df.select(from_json(col("value"), schema).alias("data")).select("data.*")

# Fraud rule: amount > 3000
fraud_df = json_df.filter(col("amount") > 3000)

# Function to display fraud alerts
def show_fraud_alerts(batch_df, epoch_id):
    rows = batch_df.collect()
    for row in rows:
        print("\n⚠FRAUD ALERT DETECTED ⚠")
        print(f"Transaction ID : {row['transaction_id']}")
        print(f"Amount : {row['amount']}")
        print("--------------------------------")

# Write stream
query = fraud_df.writeStream \
    .outputMode("append") \
    .foreachBatch(show_fraud_alerts) \
    .start()

query.awaitTermination()
```

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark M-] To Bracket M-Q Previous ^B Back ^← Prev Word
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy ^Q Where Was M-W Next ^F Forward ^→ Next Word

admin1@bflcomp20: ~/spark-3.5.1-bin-hadoop3

admin1@bflcomp20: ~/kafka_2.13-3.6.1 | admin1@bflcomp20: ~/kafka_2.13-3.6.1 | admin1@bflcomp20: ~/kafka_2.13-3.6.1 | admin1@bflcomp20: ~ | admin1@bflcomp20: ~/spark-3.5.1-bin-had...

```
(base) admin1@bflcomp20:~$ cd ~/spark-3.5.1-bin-hadoop3
(base) admin1@bflcomp20:~/spark-3.5.1-bin-hadoop3$ bin/spark-submit fraud_detector.py
26/03/05 10:03:17 WARN Utils: Your hostname, bflcomp20 resolves to a loopback address: 127.0.1.1; using 172.16.5.214 instead (on interface enp0s31f6)
26/03/05 10:03:17 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address
26/03/05 10:03:17 INFO SparkContext: Running Spark version 3.5.1
26/03/05 10:03:17 INFO SparkContext: OS info Linux, 6.8.0-100-generic, amd64
26/03/05 10:03:17 INFO SparkContext: Java version 1.8.0_482
26/03/05 10:03:17 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
26/03/05 10:03:17 INFO ResourceUtils: ==============================================================
26/03/05 10:03:17 INFO ResourceUtils: No custom resources configured for spark.driver.
26/03/05 10:03:17 INFO ResourceUtils: ==============================================================
26/03/05 10:03:17 INFO SparkContext: Submitted application: FraudDetection
26/03/05 10:03:17 INFO ResourceProfile: Default ResourceProfile created, executor resources: Map(cores -> name: cores, amount: 1, script: , vendor: , memory -> name: memory, amount: 1024, script: , vendor
: , offHeap -> name: offHeap, amount: 0, script: , vendor: ), task resources: Map(cpus -> name: cpus, amount: 1.0)
26/03/05 10:03:17 INFO ResourceProfile: Limiting resource is cpu
26/03/05 10:03:17 INFO ResourceProfileManager: Added ResourceProfile id: 0
26/03/05 10:03:17 INFO SecurityManager: Changing view acls to: admin1
26/03/05 10:03:17 INFO SecurityManager: Changing modify acls to: admin1
26/03/05 10:03:17 INFO SecurityManager: Changing view acls groups to:
26/03/05 10:03:17 INFO SecurityManager: Changing modify acls groups to:
26/03/05 10:03:17 INFO SecurityManager: SecurityManager: authentication disabled; ui acls disabled; users with view permissions: admin1; groups with view permissions: EMPTY; users with modify permissions:
 admin1; groups with modify permissions: EMPTY
26/03/05 10:03:17 INFO Utils: Successfully started service 'sparkDriver' on port 38437.
26/03/05 10:03:17 INFO SparkEnv: Registering MapOutputTracker
26/03/05 10:03:17 INFO SparkEnv: Registering BlockManagerMaster
26/03/05 10:03:17 INFO BlockManagerMasterEndpoint: Using org.apache.spark.storage.DefaultTopologyMapper for getting topology information
26/03/05 10:03:17 INFO BlockManagerMasterEndpoint: BlockManagerMasterEndpoint up
26/03/05 10:03:17 INFO SparkEnv: Registering BlockManagerMasterHeartbeat
26/03/05 10:03:17 INFO DiskBlockManager: Created local directory at /tmp/blockmgr-8492f6d3-8448-49c0-b8a9-dc9c93550a64
26/03/05 10:03:17 INFO MemoryStore: MemoryStore started with capacity 366.3 MiB
26/03/05 10:03:17 INFO SparkEnv: Registering OutputCommitCoordinator
26/03/05 10:03:17 INFO JettyUtils: Start Jetty 0.0.0.0:4040 for SparkUI
26/03/05 10:03:17 INFO Utils: Successfully started service 'SparkUI' on port 4040.
26/03/05 10:03:18 INFO Executor: Starting executor ID driver on host 172.16.5.214
26/03/05 10:03:18 INFO Executor: OS info Linux, 6.8.0-100-generic, amd64
26/03/05 10:03:18 INFO Executor: Java version 1.8.0_482
26/03/05 10:03:18 INFO Executor: Starting executor with user classpath (userClassPathFirst = false): ''
26/03/05 10:03:18 INFO Executor: Created or updated repl class loader org.apache.spark.util.MutableURLClassLoader@4f3f4e75 for default.
26/03/05 10:03:18 INFO Utils: Successfully started service 'org.apache.spark.network.netty.NettyBlockTransferService' on port 35347.
26/03/05 10:03:18 INFO NettyBlockTransferService: Server created on 172.16.5.214:35347
26/03/05 10:03:18 INFO BlockManager: Using org.apache.spark.storage.RandomBlockReplicationPolicy for block replication policy
26/03/05 10:03:18 INFO BlockManagerMaster: Registering BlockManager BlockManagerId(driver, 172.16.5.214, 35347, None)
26/03/05 10:03:18 INFO BlockManagerMasterEndpoint: Registering block manager 172.16.5.214:35347 with 366.3 MiB RAM, BlockManagerId(driver, 172.16.5.214, 35347, None)
26/03/05 10:03:18 INFO BlockManagerMaster: Registered BlockManager BlockManagerId(driver, 172.16.5.214, 35347, None)
26/03/05 10:03:18 INFO BlockManager: Initialized BlockManager: BlockManagerId(driver, 172.16.5.214, 35347, None)
26/03/05 10:03:18 INFO SharedState: Setting hive.metastore.warehouse.dir ('null') to the value of spark.sql.warehouse.dir.
26/03/05 10:03:18 INFO SharedState: Warehouse path is 'file:/home/admin1/spark-3.5.1-bin-hadoop3/spark-warehouse'.
Traceback (most recent call last):
  File "/home/admin1/spark-3.5.1-bin-hadoop3/fraud_detector.py", line 17, in <module>
    df = spark.readStream \
  File "/home/admin1/spark-3.5.1-bin-hadoop3/python/lib/pyspark.zip/pyspark/sql/streaming/readwriter.py", line 304, in load
  File "/home/admin1/spark-3.5.1-bin-hadoop3/python/lib/py4j-0.10.9.7-src.zip/py4j/java_gateway.py", line 1322, in __call__
  File "/home/admin1/spark-3.5.1-bin-hadoop3/python/lib/pyspark.zip/pyspark/errors/exceptions/captured.py", line 185, in deco
```

admin1@bflcomp20: ~/spark-3.5.1-bin-hadoop3

admin1@bflcomp20: ~/kafka_2.13-3.6.1 | admin1@bflcomp20: ~/kafka_2.13-3.6.1 | admin1@bflcomp20: ~/kafka_2.13-3.6.1 | admin1@bflcomp20: ~ | admin1@bflcomp20: ~/spark-3.5.1-bin-had...

```
Amount : 4388
------------------------------

⚠ FRAUD ALERT DETECTED ⚠
Transaction ID : 9713
Amount : 3839
------------------------------

⚠ FRAUD ALERT DETECTED ⚠
Transaction ID : 7385
Amount : 4922
------------------------------

⚠ FRAUD ALERT DETECTED ⚠
Transaction ID : 8488
Amount : 4860
------------------------------

⚠ FRAUD ALERT DETECTED ⚠
Transaction ID : 3652
Amount : 3985
------------------------------

⚠ FRAUD ALERT DETECTED ⚠
Transaction ID : 5171
Amount : 4251
------------------------------

⚠ FRAUD ALERT DETECTED ⚠
Transaction ID : 8562
Amount : 4708
------------------------------

⚠ FRAUD ALERT DETECTED ⚠
Transaction ID : 5174
Amount : 3686
------------------------------

⚠ FRAUD ALERT DETECTED ⚠
Transaction ID : 5220
Amount : 4689
------------------------------

⚠ FRAUD ALERT DETECTED ⚠
Transaction ID : 4936
Amount : 3913
------------------------------

⚠ FRAUD ALERT DETECTED ⚠
Transaction ID : 4590
Amount : 4479
------------------------------

⚠ FRAUD ALERT DETECTED ⚠
```